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SEQUENCE & SELECTION

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Sequence example

```
DECLARATION:
    M, P    : integer

ALGORITHM:
    M ← 16
    P ← M * 2
    M ← P + 100
    P ← P + M
```

M = 132 and P = 164



Sequence

- This Algorithm are sequences of one or more instructions.
- It means:
  - Each instruction is done one by one;
  - Each instruction is carried out exactly once;
  - no instructions are repeated;
  - The sequence of instructions carried out by the processor is the same as the sequence of instructions as written in the text of the algorithm;
  - The end of the last instruction is the end of the algorithm.



effect of the order of instructions

```
1   PROGRAM SEQUENCE_1
2   ( example of sequence algorithm 1 )
3
4   DECLARATION:
5       A, B : integer              { variable }
6
7   ALGORITHM:

8       A ← 10
9       A ← 2 * A
10      B ← A
11      Write (B)
```

```
1   PROGRAM SEQUENCE_2
2   ( example of sequence algorithm 2 )
3
4   DECLARATION:
5       A, B : integer              { variable }
6
7   ALGORITHM:

8       A ← 10
9       B ← A
10      A ← 2 * A
11      Write (B)
```



Sequence examples

```
1   PROGRAM HELLO_WORLD1
2   (algorithm to print "Hello, world"  )
3
4   DECLARATION:
5       { none }
6
7   ALGORITHM:

8       Write ('Hello, World')
```

```
1   PROGRAM HELLO_WORLD3
2   (algorithm to print "Hello, world"  )
3
4   DECLARATION:
5       Const message = 'Hello, world'
6
7   ALGORITHM:

8       Write (message)
```

```
1   PROGRAM HELLO_WORLD2
2   (algorithm to print "Hello, world"  )
3
4   DECLARATION:
5       message : string   { sentence variable to be printed }
6
7   ALGORITHM:

8       message ← 'Hello, World'
9       Write (message)
```



Sequence examples

```
1   PROGRAM TRIANGLE_AREA
2   (Read the base length (a) and height (t) of a triangle. Calculate
3   the area of the triangle. The formula L = ½ at. The L value is
4   printed to the output device.)
5
6   DECLARATION:
7       a: real       {length of the triangle base, in cm}
8       t: real       {height of triangle, in cm}
9       L: real       {area of triangle, in cm²}
10
11  ALGORITHM:

12      read (a)
13      read (t)
14      L ← a * t / 2.0
15      Write (L)
```



Sequence practice:

```
PROGRAM EmployeeNetSalary
{ Calculate the employee's net salary. Data input is the name of the
employee and his monthly base salary. Net salary = base salary + benefits -
tax. Benefits are 20% of basic salary, while taxes are 15% of basic salary
and benefits. Output is the name of the employee and his net salary. }
```



Sequence practice:

```
1   PROGRAM EmployeeNetSalary
2   (Calculate the employee's net salary. Data input is the name of the
3   employee and his monthly base salary. Net salary = base salary +
4   benefits - tax. Benefits are 20% of basic salary, while taxes are
5   15% of basic salary and benefits. Output is the name of the employee
6   and his net salary.)
7
8   DECLARATION:
9       const percentBenefits = 0.2   {percentage of salary benefits}
10      const percentTax = 0.15       {percentage of tax deduction}
11      employeeName: string
12      basicSalary, benefits, taxes, netSalary: real
13
14  ALGORITHM:

15      read (employeeName, basicSalary)
16      benefits ← percentBenefits * basicSalary
17      taxes ← percentTax * (basicSalary + benefits)
18      netSalary ← basicSalary + benefits - taxes
19      Write (employeeName, netSalary)
```



Sequence practice:

```
1   PROGRAM MidPoint
2   (Calculate the midpoint of two points P1 = (x1, y1) and P2 = (x2,
3   y2). The midpoint is written to the output device. Notice that point
4   is a structured data type in the form of records.)
5
6   DECLARATION:
7       type Point : record
8           <    x : real,
9                y : real
10          >
11
12      P1, P2, midPoint : Point
13
14  ALGORITHM:

15      read (P1.x, P1.y)                { P1 point}
16      read (P2.x, P2.y)                { P2 point}
17      midPoint.x ← (P1.x + P2.x) / 2.0
18      midPoint.y ← (P1.y + P2.y) / 2.0
19      Write (midPoint.x, midPoint.y)
```



Selection of one case

```
if condition then
      statement(s)
endif
```

```
if x > 100 then
      x ← x + 1
endif
```

```
if kar = '*' then
      stop ← true
endif
```

```
if (a = 0) or (p = 1) then
      q ← a * P
      write(q)
endif
```

```
if exist = false then
      read (cc)
      write (cc)
endif
```

Could be written

```
if not exist then
      read (cc)
      write (cc)
endif
```



Selection of one case

```
1   PROGRAM EvenNumber
2   (Prints an "even number" message if an integer that is read from the
3   input device is an even number.)
4
5   DECLARATION:
6         x : integer                    {The number to be determined}
7   ALGORITHM:
8       read (x)
9       if (x mod 2) = 0 then
10        write('even number')
11      endif
```



Selection of one case

main.c

```c
#include <stdio.h>

int main(void) {
  //Program Even Number
  //Prints an "even number" message if an integer that is read from the input device is an even number

  //Declaration
  int x;

  //Algorithm
  scanf( format: "%d",&x);
  if(x%2==0) {
    printf( format: "even number");
  }
}

```



Selection of two cases

```
if a > 0 then
      write('positive number')
else
      write('not a positive number')
endif
```

```
if condition then
      statement(s) 1
else
      statement(s) 2
endif
```

```
if (k > 4) and (k div 2 = 4) then
      read(n)
      z ← n * k
else
      read(m)
      z ← m / k
endif
```



Selection of two cases, ex:

```
1   PROGRAM EvenOrOdd
2   (Prints an "even" message if an integer read from the input device
3   is an even number, or "odd" if it is not even.)
4
5   DECLARATION:
6         x : integer                    {The number to be determined}
7   ALGORITHM:
8       read (x)
9       if (x mod 2) = 0 then
10        write('even')
11      else
12        write('odd')
13      endif
```



Selection of two cases, ex:

main.c

```c
#include <stdio.h>

int main(void) {
  //Program Even Number
  /*Prints an "even number" message if an integer that is read from the input device is an even number
  and prints an "odd number: message if an integer that is read from the input device is an odd number*/

  //Declaration
  int x;

  //Algorithm
  scanf( format: "%d",&x);
  if(x%2==0) {
    printf( format: "even number");
  }else {
    printf( format: "odd number");
  }
}
```



Selection of thre case or more

```
if condition 1 then
      statement(s) 1
else
      if condition 2 then
            statement(s) 2
      else
            if condition 3 then
                  statement(s) 3
            endif
      endif
endif
```



Selection of thre case or more, ex:

```
1   PROGRAM NumbersType
2   (Prints a "positive number" message if an integer read from the
3   input device > 0, or "negative number" if the number is < 0, or
4   "zero" if the number = 0.)
5
6   DECLARATION:
7         x : integer                    {The number to be determined}
8   ALGORITHM:
9         read (x)
10        if x > 0 then
11              write('positive number')
12        else
13              if x < 0 then
14                    write('negative number')
15              else
16                    if x = 0 then
17                          write('zero')
18                    endif
19              endif
20        endif
```

```
9         read (x)
10        if x > 0 then
11              write('positive number')
12        else
13              if x < 0 then
14                    write('negative number')
15              else
16                    write('zero')
17              endif
18        endif
```



Selection of thre case or more, ex:

main.c

```c
#include <stdio.h>

int main(void) {
  //Program Number Type

  //Declaration
  int x;

  //Algorithm
  scanf( format: "%d",&x);
  if(x>0) {
    printf( format: "positive number");
  }else if(x<0) {
    printf( format: "negative number");
  }else if(x==0) {
    printf( format: "zero");
  }
}
```



CASE construction

```
case (expression)
      <value1>   : statement1
      <value2>   : statement2
      <value3>   : statement3
      .
      .
      .
      <valueN>   : statementN
      [otherwise statementX]
endcase
```

the expression is any expression (arithmetic or boolean) that produces a value.



CASE construction, ex:

```
1    PROGRAM EvenOrOdd
2    (Prints an "even" message if an integer read from the input device
3    is an even number, or "odd" if it is not even.)
4
5    DECLARATION:
6          x : integer                    {The number to be determined}
7    ALGORITHM:
8        read (x)
9        case (x mod 2)
10         0 : write('even')
11         1 : write('odd')
12       endcase
```



CASE construction, ex:

```
1   PROGRAM EvenOrOdd
2   (Prints an "even" message if an integer read from the input device
3   is an even number, or "odd" if it is not even.)
4
5   DECLARATION:
6         x : integer                    {The number to be determined}
7   ALGORITHM:
8       read (x)
9       case (x mod 2)
10        0 : write('even')
11        1 : write('odd')
12      endcase
```



CASE construction, ex:

main.c

```c
#include <stdio.h>

int main(void) {
  //Program Even Number
  /*Prints an "even number" message if an integer that is read from the input device is an even number
  and prints an "odd number: message if an integer that is read from the input device is an odd number*/

  //Declaration
  int x;

  //Algorithm
  scanf( format: "%d",&x);
  switch (x%2) {
    case 0: printf( format: "even");
    break;
    case 1: printf( format: "odd");
    break;
  }
}
```



CASE construction, ex:

```
1   PROGRAM MonthName
2   (Print month's name based on month number (1..12).)
3
4   DECLARATION:
5         monthNumber : integer        {Month number to be determined}
6   ALGORITHM:
7       read (monthNumber)
8       case (monthNumber)
9         1     : write('January')
10        2     : write('February')
11        3     : write('March')
12        4     : write('April')
13        5     : write('May')
14        6     : write('June')
15        7     : write('July')
16        8     : write('August')
17        9     : write('September')
18        10    : write('October')
19        11    : write('November')
20        12    : write('December')
21        otherwise: write('incorrect number')
22      endcase
```



CASE construction, ex:

main.c

```c
int main(void) {
  //MonthName
  //print month's name based on month number (1..12)

  //Declaration
  int monthNumber;

  //Algorithm
  scanf( format: "%d",&monthNumber);
  switch (monthNumber) {
    case 1: printf( format: "January"); break;
    case 2: printf( format: "February"); break;
    case 3: printf( format: "March"); break;
    case 4: printf( format: "April"); break;
    case 5: printf( format: "June"); break;
    case 6: printf( format: "July"); break;
    case 7: printf( format: "August"); break;
    case 8: printf( format: "September"); break;
    case 9: printf( format: "October"); break;
    case 10: printf( format: "November"); break;
    case 11: printf( format: "December"); break;
    case 12: printf( format: "December"); break;
    default: printf( format: "Incorrect Number");break;
  }
}
```



CASE construction, ex:

```
28      //Declaration
29      int x;
30
31      scanf( format: "%d",&x);
32      if(x>0) {
33          printf( format: "Positive Number");
34          if(x<0) {
35              printf( format: "Negative Number");
36          }
37      }
38  }
```



CASE construction, ex:

```
28      //Declaration
29      float x;
30
31      scanf( format: "%f",&x);
32      if(x>0.0) {
33        printf( format: "Positive Number");
34        if(x<10.0) {
35          printf( format: "The number is smaller than 10");
36        }else if(x==10) {
37          printf( format: "The number is  10");
38        }
39        else{
40          printf( format: "The number is greater than 10");
41        }
42      }
43  }
```



Bibliography

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Thank You!



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